

Master Theorem

• For a general recurrence $T(n) = aT(n/b) + f(n)$ where $f(n) \in O(n^d)$

- If $a < b^d \Rightarrow T(n) \in O(n^d)$
- If $a = b^d \Rightarrow T(n) \in O(n^d \log n)$
- If $a > b^d \Rightarrow T(n) \in O(n^{\log_b a})$

Algo - Lab 06

1. Solve the following recurrences using the Master Theorem.

4.1 $T(n) = 4T(n/2) + n, T(1) = 1$

• $a = 4$

• $b = 2$

• $f(n) = n$

* Step 2 Compute the critical term

$$n^{\log_b a} = n^{\log_2 4}$$

$$n^{\log_2 4} = n^2$$

* Step 3 Compare $f(n)$ with $n^{\log_b a}$

• $f(n) = n$

• $n^{\log_b a} = n^2$

$$n = O(n^{2-1}) \Rightarrow f(n) = O(n^{\log_b a - \epsilon})$$

* So:

$$T(n) = \Theta(n^{\log_b a}) = \Theta(n^2)$$

$$T(n) = \Theta(n^2)$$

4.2 $T(n) = 4T(n/2) + n^2, T(1) = 1$
 • Master theorem form

• $a = 4$

• $b = 2$

• $f(n) = n^2$

• Critical term:
 $n \log_b^a = n \log_2^4 = n^2$

• Compare $f(n)$ with $n \log_b^a$

• $f(n) = n^2$

• $n \log_b^a = n^2$

• $f(n) = \Theta(n \log_b^a)$

$T(n) = \Theta(n \log_b^a \log n) = \Theta(n^2 \log n)$

• $T(n) = \Theta(n^2 \log n)$

4.3 $T(n) = 4T(n/2) + n^3, T(1) = 1$

$a = 4$

$b = 2$

$f(n) = n^3$

• Critical term
 $n \log_b^a = n \log_2^4 = n^2$

• Compare $f(n)$ with $n \log_b^a = n^2$, since $n^2 < n^3$, then:

$T(n) = O(n^3)$

• $a f(n/b) \leq c f(n), \forall c < 1$

$= 4 \left(f\left(\frac{n}{2}\right) \right) = 4 \left(\frac{n}{2} \right)^3 = \frac{n^3}{2} \leq c n^3$ // complies